VIPHER AID - ANIMAL RESCUE APP

♦INTRODUCTION :

Animal rescue organization around the world faces many challenges while trying to help injured and abandoned street animals. Even though there are many dedicated NGO's, veterinarians and volunteers, the biggest problem is coordination. Most rescue information is shared through WhatsApp groups, or random social media posts. While this works in urgent situations, it often leads to confusion, repeated reports, missed calls and no proper tracking of what happened next.

One of the main issues is lack of a centralised platform where all rescue-related information can be stored and monitored. When reports are scattered across different communication channels, there is no proper record of cases, no transparency in updates and no way for citizens to know whether an animal has been rescued or not. Overtime, this reduces efficiency and trust in the rescue system.

Previous system such as Stray Rescue Management System (SRMS) developed by A/P Suburamaniyam and Kutty Mammi [1] and the web-based shelter coordinating system proposed by Wee et al [2] , show that digital platforms can improve animal rescue management. These system focus on structured reporting, donation transparency and shelter coordination. However many existing solutions are either complex, region specified or a require larger infrastructure setups.

To address these practical challenges, this project introduces Vipher Aid , a simple and lightweight web-based street animal rescue platform. The main aim of this system is to bring all rescue stakeholder into one centralized digital environment. Citizens can report injured or distressed animals through an online form by providing details such as animal type, location, urgency level, description and image evidence. These reports are stored in a database and displayed on a secure dashboard.

Authorized rescue personnel, including NGO members and veterinary professionals, access the system through a unified login system. From the dashboard they can view active cases and update the status of rescue operations. This structured workflow reduces miscommunication and ensures that rescue cases are not lost or forgotten.

Vipher Aid focusses on simplicity, accessibility and practical implementation. It does not require complex server infrastructure and can be deployed easily for small or medium scale rescue organizations. By centralizing communication and maintain digital

records the system improves accountability, coordination and overall efficiency in street animal rescue operations.

♦ CITATION:

1. A/P Subramaniam, K., & Kutty Mammi, H. (2025). Stray Rescue Management System (SRMS). International Journal of Innovative Computing, 15(1), 139–144. https://doi.org/10.11113/ijic.v15n1.501
2. 2.Wee, M., Hafid, H., & Bernard Leong. (2022). Development of Web-based system for Animal Shelter and Rescue in Johor State. Applied Information Technology And Computer Science, 3(2), 479-495. https://publisher.uthm.edu.my/periodicals/index.php/aitcs/article/view/7645

♦ BIBTEXT :

1. @article {mishra_animal_monitoring, author = {Arvind Kumar Mishra and others}, title = {Service-Oriented Architecture for Animal Monitoring System}, journal = {Revista Electronics de Veterinarian}, year = {2019} }
2. @article {jagtap_shelter_soul, author = {Y. D. Jagtap}, title = {Shelter Soul: Bridging Shelters and Adopters Through Technology}, journal = {arXiv preprint arXiv:2506.12739}, year = {2025}}

♦ PROBLEM STATEMENT :

Street animal rescue operations often face several challenges due to the lack of a proper coordination system. Currently, most rescue reports are shared through informal communication platforms such as WhatsApp groups and social media posts. While these platforms allow quick sharing of information, they do not provide an organized way to manage rescue cases.

As a result, multiple volunteers may respond to the same rescue request, causing duplication of efforts. As the same time, some rescue cases may be ignored or delayed because important messages get buried among numerous conversation.

Another major issue is the absence of a centralized database to track rescue operations. Without proper records, it becomes difficult for organizations to monitor rescue activities, manage shelter availability or track donations used for animal care. These limitations highlight the need for a centralized digital platform that can efficiently manage rescue reports and improve coordination among rescue stakeholders.

♦ OBJECTIVES :

The primary objective of the Vipher Aid system is to develop a centralized web platform that improves the efficiency of street animal rescue operations.

The main objectives include :

- To provide an online platform where citizens can report injured or distressed animals.
- To create a centralized database for storin and tracking rescue cases.
- To provide a dashboard for rescue organization to manage rescue activities.
- To enable tracking of rescued animals and shelter availability.
- To manage donations and funds used for animal rescue and medical treatment.
- To improve coordination between citizens, volunteers, NGOs and veterinarians.

♦ LITERATURE REVIEW :

Several digital systems have been developed to improve animal recue and shelter management.

- The Stray Rescue Management System (SRMS) developed by A/P Subramaniyam and Kutty Mammi provides a structured systems for recording and managing Stray animal rescue operations. This system focuses on maintaining records of rescued animals and improving internal rescue management.
- Another system proposed by M.Wee,Hafid H., and Bernard Leonng introduces a web-based platform designed to coordinate animal shelters and rescue organizations. The system focuses mainly on shelter management and adoption processes.
- Research by Aravind Kumar Mishra proposes a service-oriented architecture for monitoring animal-related activities using digital technologies.
- Similarly, Y.D. Jagtap introduced the Shelter Soul Platform, which connects animal shelters with potential adopters through digital platforms.

  Although these systems improve certain aspects of animal welfare management, many of them focus primarily on shelter operations or adoption management rather than real time rescue reporting. The Vipher Ais system aims to address this gap by provide a centralized platform for real-time rescue reporting and management.

♦ COMPARISION WITH EXISTING SYSTEMS :

The provided Vipher Aid system improves several aspects of animal rescue management when compared with the Stray Rescue Management System (SRMS) development by A/P Subramaniyam and Kutty Mammi.While SRMS mainly focuses on managing rescue operations within organizations,Vipher Aid expands the system by allowing public participation and centralized case tracking.

Feature	SRMS	Vipher Aid
Reporting Method	Rescue reports created by NGO staff	Citizens can directly report injured animals
Accessibility	Mainly used within organizations	Accessible to citizens, NGOs, and volunteers
Case Tracking	Basic record management	Real-time rescue status updates
Dashboard Modules	Limited management features	Report, Rescue, Shelter, and Fund modules
Coordination	Focused on internal rescue teams	Connects citizens, NGOs, and volunteers
System Complexity	Requires structured setup	Lightweight web-based platform

From the comparison, it can be observed that SRMS mainly supports internal rescue management, whereas Vipher Aid provides a more open and interactive platform. By allowing citizens to submit rescue reports and by integrating multiple dashboard modules such as reporting, rescue tracking, shelter management and fund monitoring, Vipher Aid improves coordination and transparency in animal rescue operations.

- **SYSTEM ARCHITECHTURE**:

The Vipher Aid system follows a three-layer architecture consisting of the user interface layer, application layer and database layer.

- **APPLICATION LAYER:**

The application layer processes user requests and manages system operations. It connects the user interface with the database and handles authentication, case management and status updates.

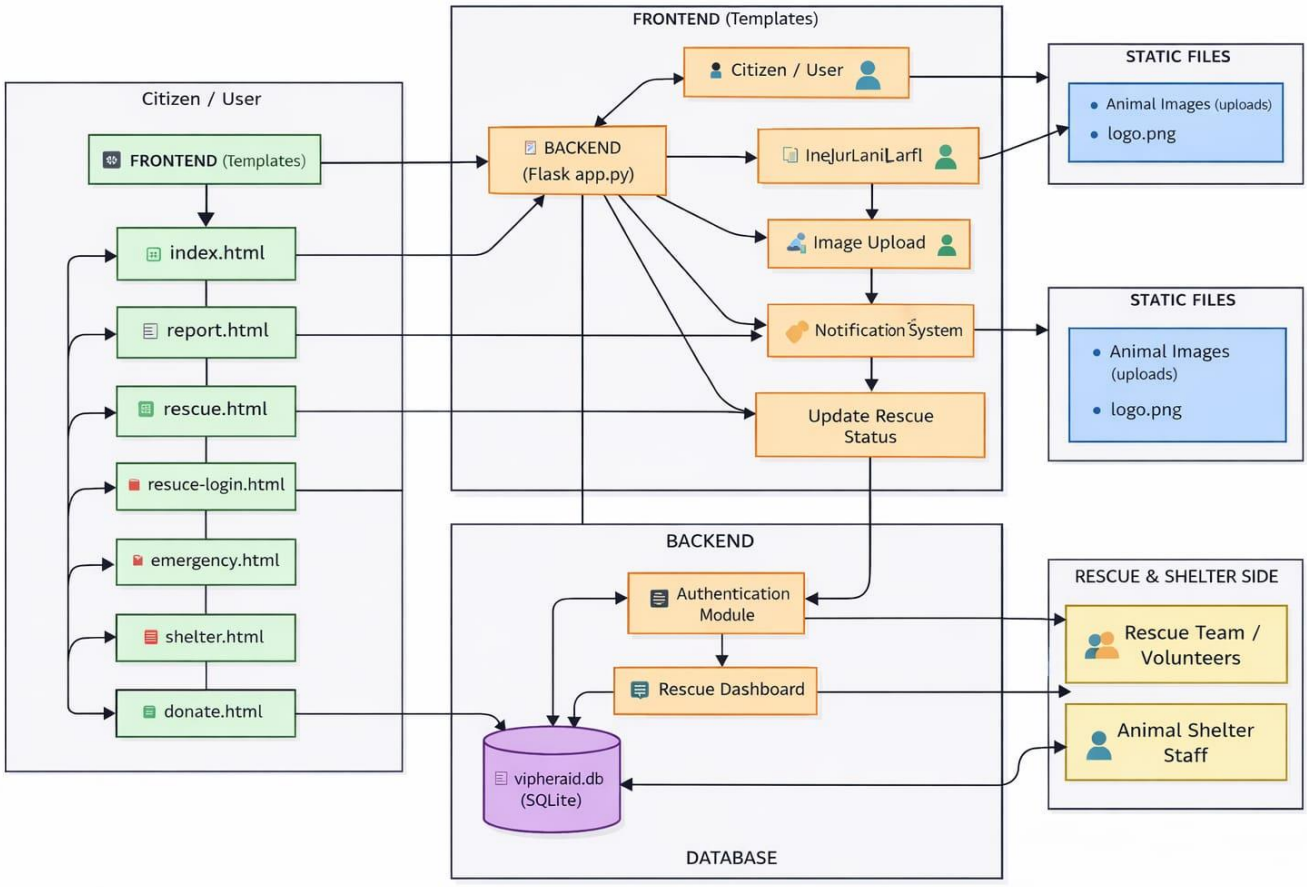
➢ DATABASE LAYER:

The database layer stores rescue, reports, shelter details, donation records and user information.

➢ DASHBOARD MODULES:

The system includes a management dashboard with several modules:

- Report Module: Display rescue reports submitted by citizens.
- Rescue Modules: Allows rescue teams to manage and update rescue operations.
- Shelter Modules: Maintains information about available shelters for rescued animals.
- Fund Modules: Tracks donations and financial support for rescue activities.
- Emergency Modules: It maintains the call records of the report.



♦ METHODOLOGY:

The vipher Aid platform is designed as a web-based that enables citizens to report injured street animals and allows rescue organisation to manage rescue operations efficiently. The methodology describes the architecture of the system, frontend and backend implementation, database structure, APIs, security mechanisms and enhanced features such as emergency contact and transparent funding.

➢ OVER ALL ARCHITECHTURE :

The vipher Aid system follows a three-layer architecture consisting of the presentation layer, application layer and database layer.

1. PRESENTATION LAYER:

This layer provides the user interface for citizens and rescue organizations. Citizens can submit rescue reports through an online form, while authorized users access the dashboard to manage rescue operations . This system includes the GPS based location input feature,which helps in accurately identifying the rescue location.An Emergency Hotline button is available on the homepage,allowing users to directly contact rescue teams for urgent situations.

2.APPLICATION LAYER:

The application layer processes user requests and performs system operations such as:

- Manual rescue report validation and storage.
- User authentication and authorization.
- Rescue case status updates.
- GPS location processing for accurate tracking .
- Emergency call request handling.
- Shelter and donation management.

3.DATABASE LAYER:

The Database layer stores all system – related data in an organized manner, including:

- Rescue reports submitted by citizens.
- User login and authentication details.
- Shelter information and availability .
- Donation and funding records.
- Rescue status updates.
- Emergency call records.

➢ FRONTEND DESIGN :

The frontend of the Vipher Aid platform is designed to provide a simple and accessible interface for users.

The frontend includes the following pages :

- User Login page for rescue organization.
- Rescue report submission form for citizens.
- Dashboard interface to view rescue causes.
- Shelter and fund management modules.
  The interface is developed using HTML,CSS which enables responsive web design and smooth user interactions.

➢ BACKEND SERVICES :

The backend service handle the core functionality of the system and manage communication between the forward interface and the database.

Major backend service include :

- Processing rescue report submissions.
- Managing user authentications.
- Updating rescue case status.

- Retrieving dashboard data.
- Managing shelter and donation information.

The backend is implemented using python with Flask framework, which allows lightweight and efficient web application development.

➢ DATABASE SCHEMA :

The main database tables include :

TABLES	DESCRIPTION
USER	Stores login credentials for administrators and rescue teams
REPORT RESCUE	Stores information submitted by citizens about injured animals
SHELTER	Maintains details of animal shelters and availability
DONATION	Tracks funds donated for rescue operations
RESCUE	Records updates related to rescue activities
EMERGENCY – CALL	Stores emergency contact request records.

➢ APIs :

Application Programming Interface (APIs) are used to enable communication between the frontend and backend service.

Key APIs in the system includes:

- Report Submission API : Receives rescue reports from the frontend and stores them in database.
- Login API : Authenticates authorized rescue personnel.
- Rescue Update API : Updates the status of rescue operations.
- Dashboard API : Retrieves rescue case information for display.

These APIs ensure efficient data transfer and system functionality.

♦ SECURITY MECHANISMS :

To ensure data security and controlled system access, several security mechanisms are implemented.

These include:

- User authentication through login credentials.
- Role-based access control for authorized rescue organizations.
- Input validation for rescue report forms.
- Secure database storage for user and rescue data.

These mechanisms help protect the system from unauthorized access and ensure data integrity.

♦ TECHNOLOGY STACK :

The Vipher Aid platform is developed using the following technologies:

LAYER TECHNOLOGY

- Frontend : HTML , CSS , JAVASCRIPT
- Backend : python , Flask
- Database : SQlite
- Development Environment Virtual
- Environment (python)
- Tools : Visual Studio Code

This technology stack enables the development of a lightweight and scalable web application.

♦ IMPLEMENTATION:

The Implementation of the vipher aid platform follows a atructured rescue reporting worflow.

1. A citizen encounters a injured or distressed stray animal and submits a report through the web platform.
2. OR Use the emergency Hotline button for immediate assistance.
3. The report details, including location,animal type ,description ,and image , are stored in the database.
4. Authorized rescue organizations log in to the dashboard to view active rescue cases.

5. The rescue team reviews the report and responds to the location.
6. Once the rescue operation is completed , the rescue status is updated in the system.
7. If required, the rescued animal is transferred to a shelter for treatment and care.

All rescue activities are recorded in the database to ensure proper tracking and coordination of rescue operations.

♦ SYSTEM PERFORMANCE OBSERVATIONS:

Basic performance observations were recorded during testing.

PARAMETER	OBSERVATION
Average page load time	1.5 - 2 seconds
Api Response Time	approximately 400 – 600 ms
Concurrent Users Tested	upto 10 users
System Stability	Stable during testing

♦ SYSTEM INTERFACE SCREENSHOTS:

The following screenshots illustrate the key components of the Vipher Aid platform:

- Login page :

Interface used by rescue organizations to securely accecss the system.

- Rescue Report Form:

Page where citizens submit details of injured or distressed animals .

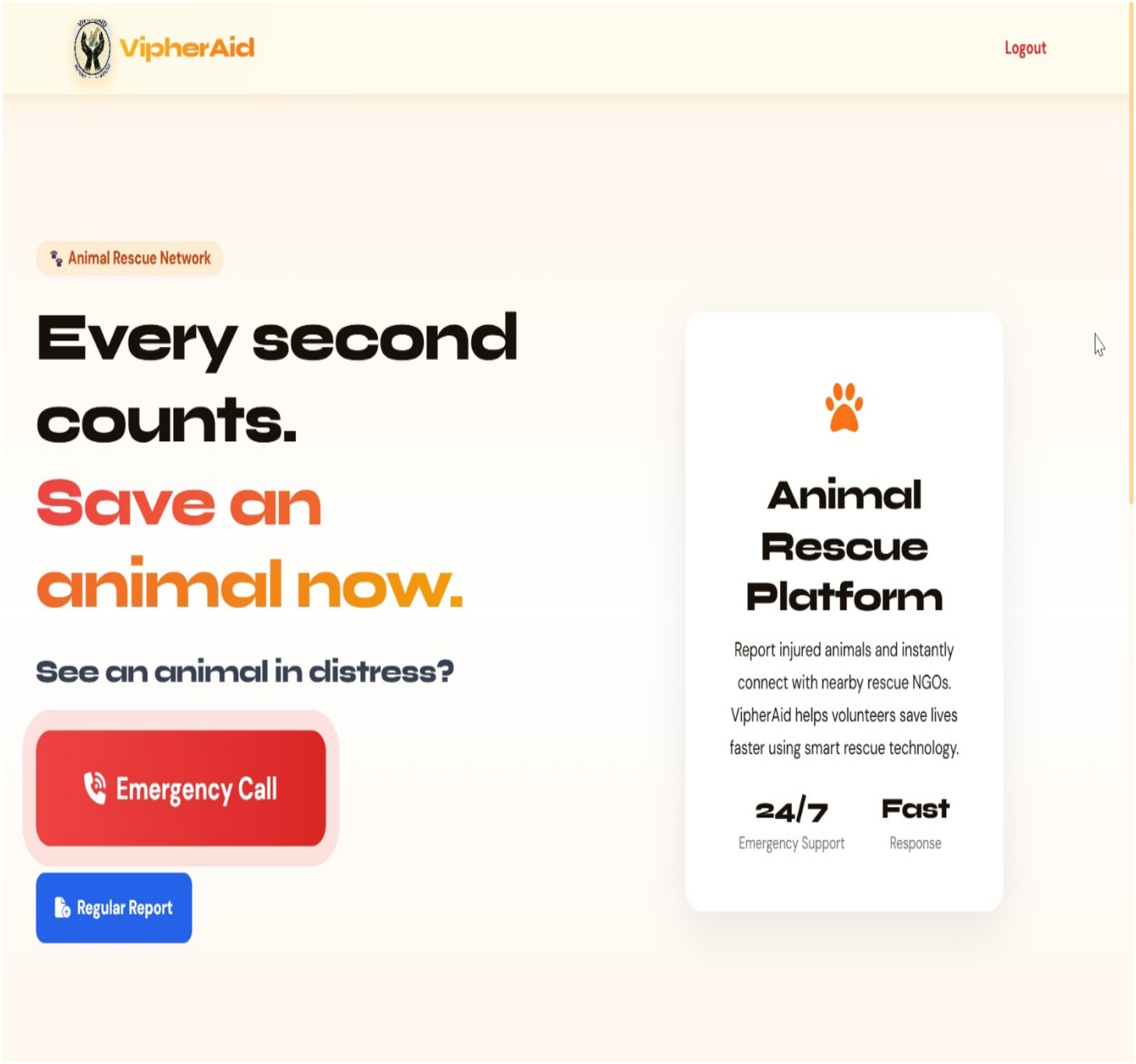
- Dashboard view :

Displays active rescue cases and allows administrators to manage rescue operations.

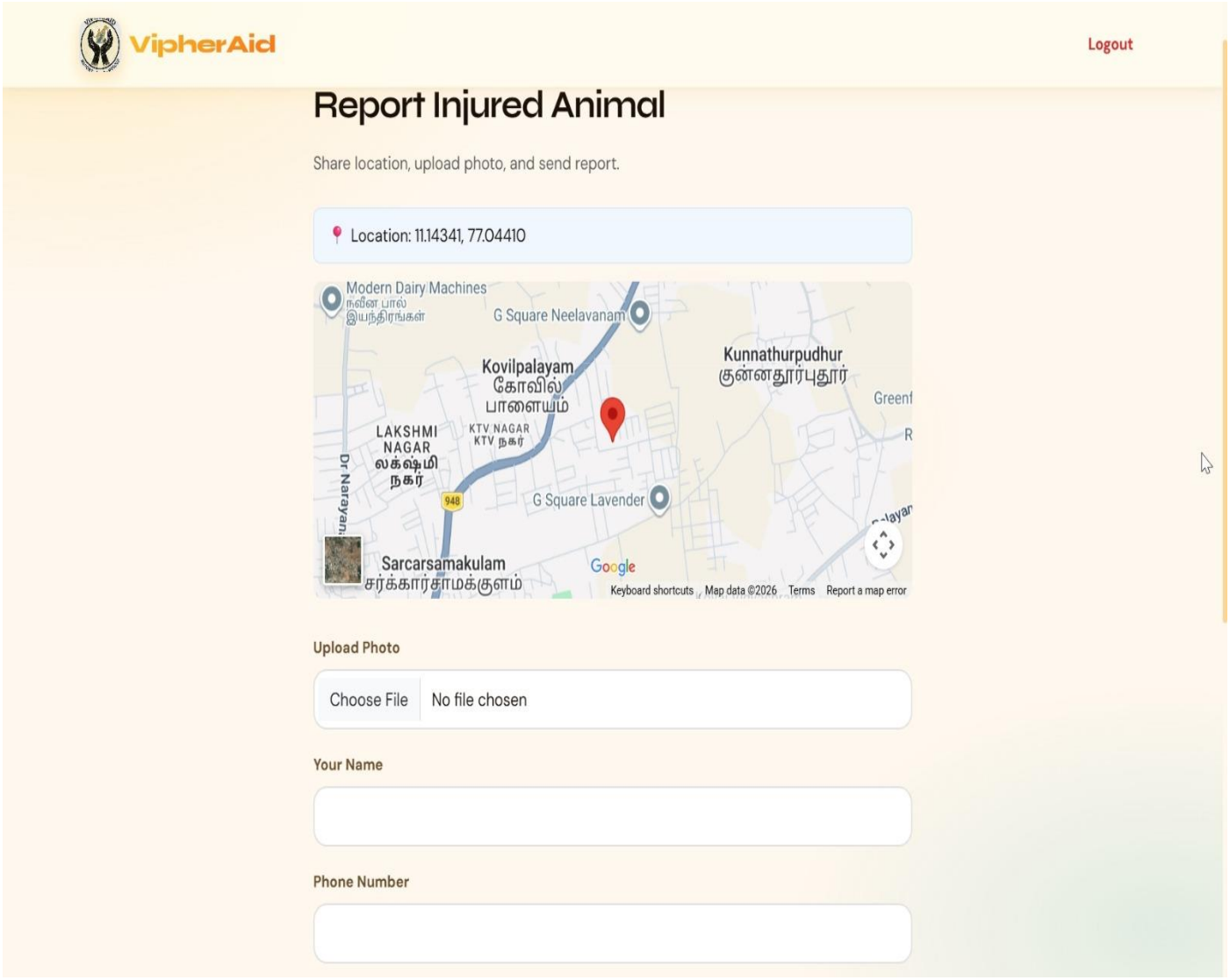
- Rescue Status Update Page:

Interface used by rescue teams to update rescue progress and case status.

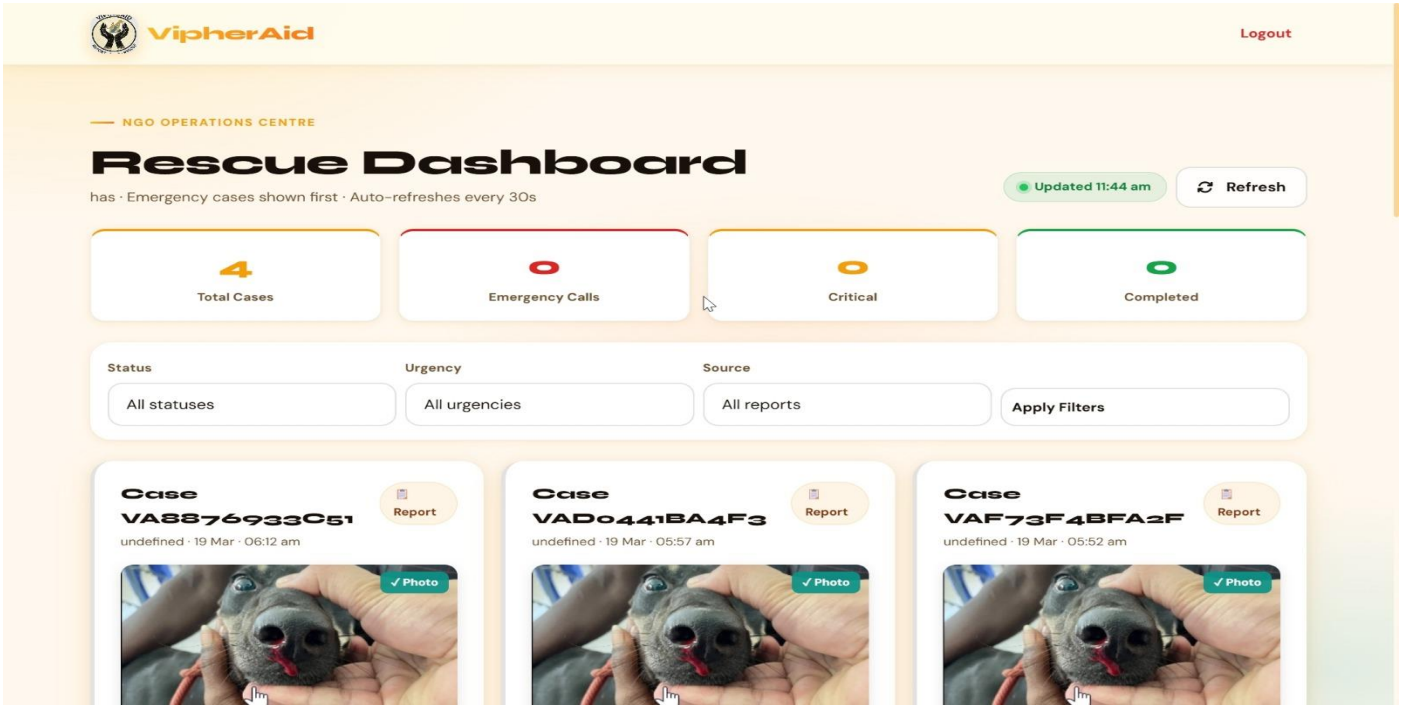
HOME PAGE


VipherAid
Logout
Animal Rescue Network
Every second counts.
Save an animal now.
See an animal in distress?
Emergency Call
Regular Report
Animal Rescue Platform
Report injured animals and instantly connect with nearby rescue NGOs. VipherAid helps volunteers save lives faster using smart rescue technology.
24/7
Emergency Support
Fast
Response

REPORT PAGE

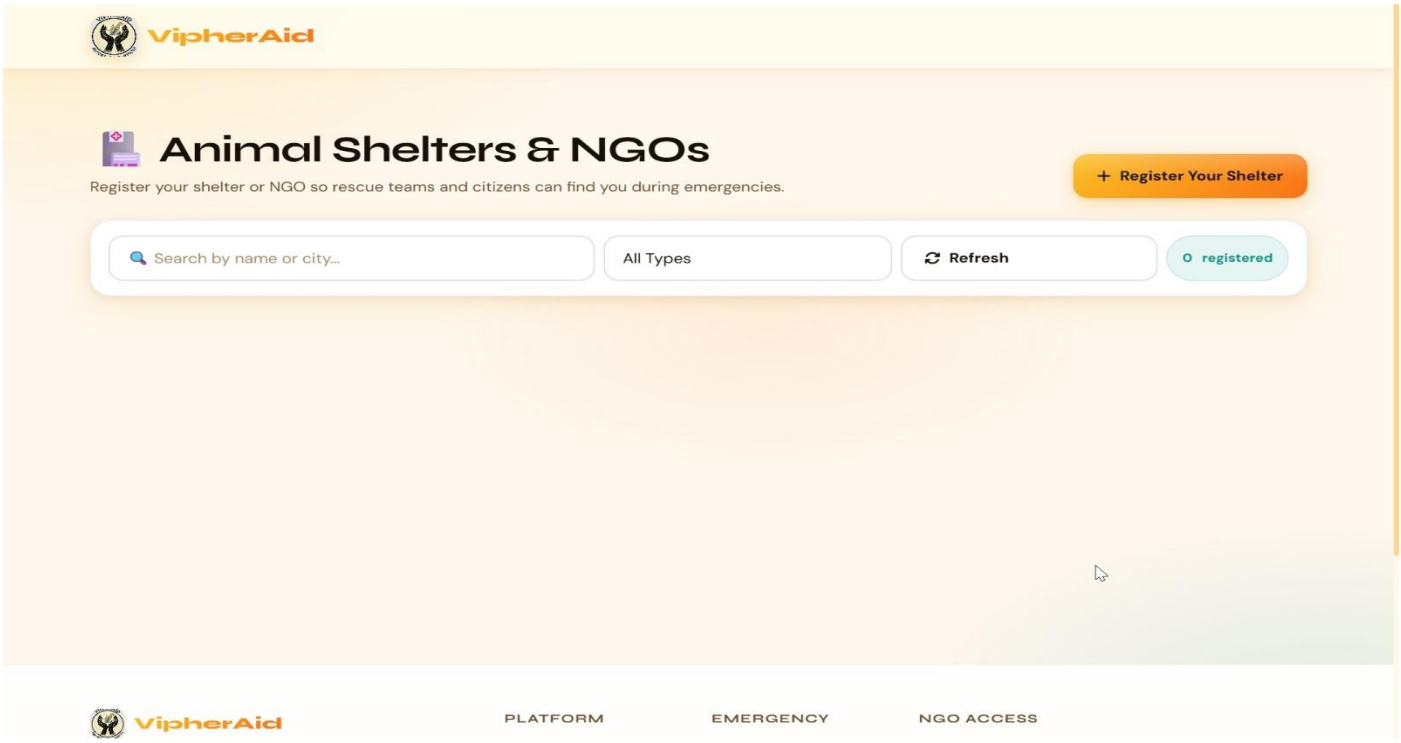


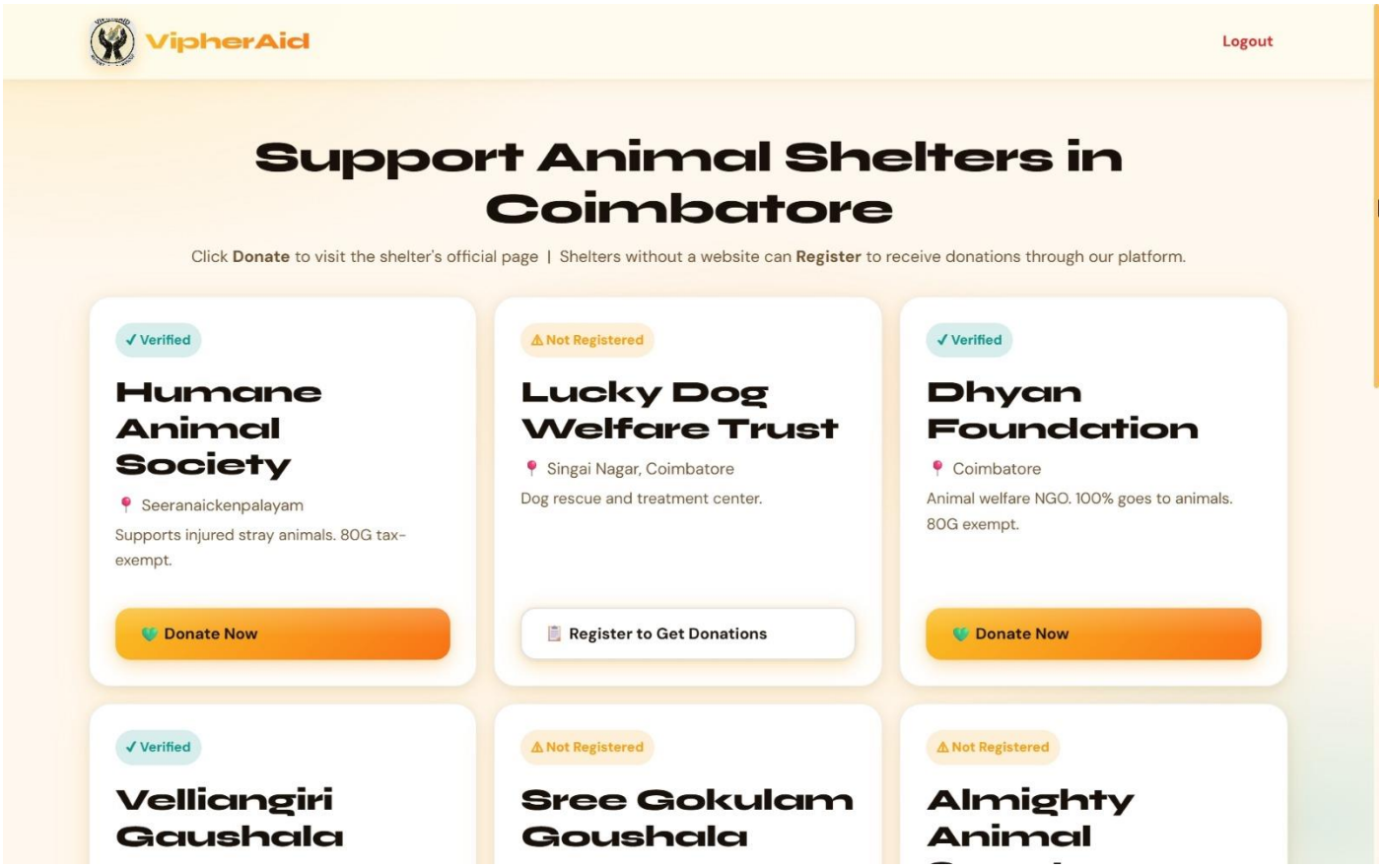
RESCUE PAGE



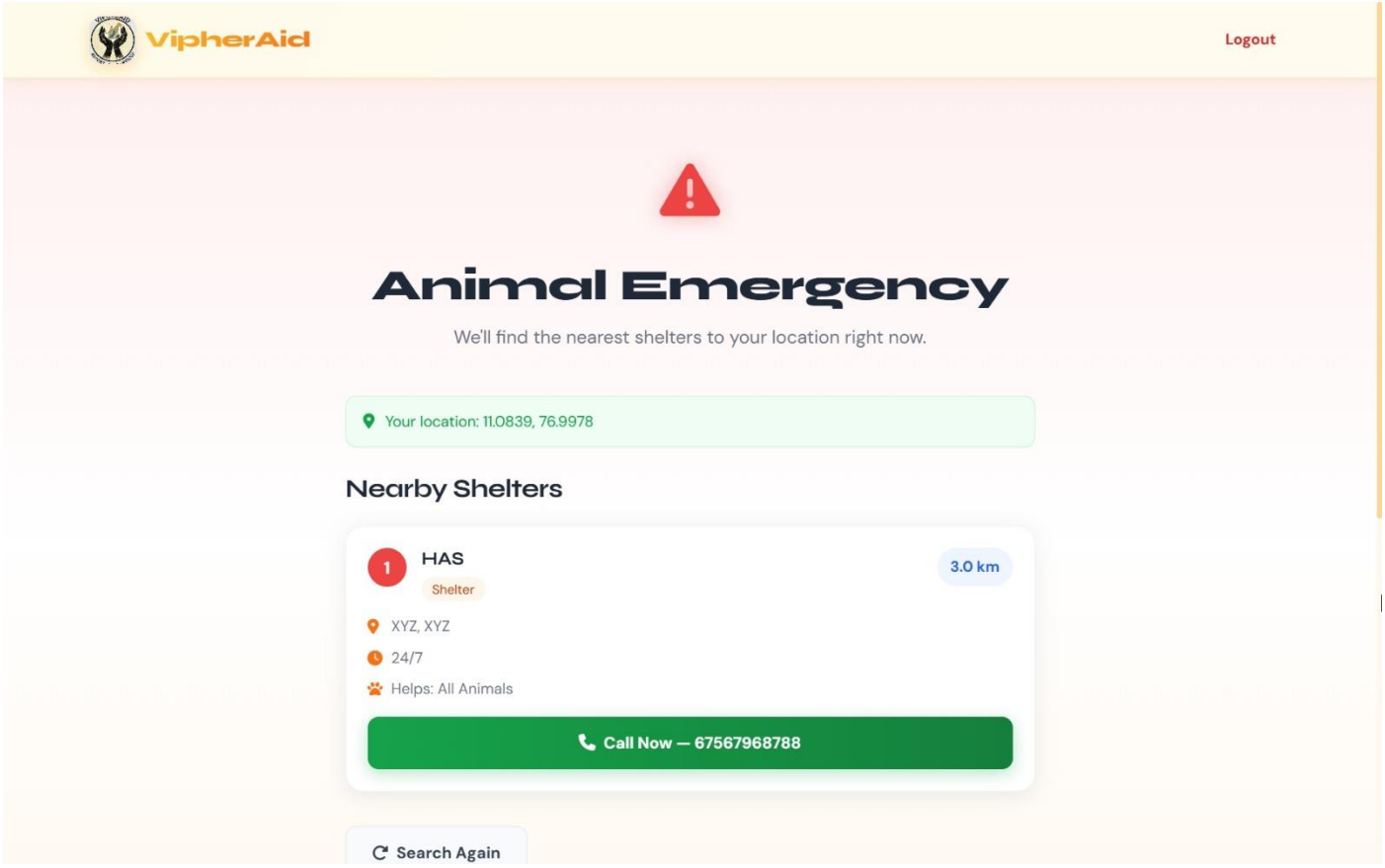


SHELTER PAGE





EMERGENCY HOTLINE


VipherAid
Logout
Animal Emergency
We'll find the nearest shelters to your location right now.
Your location: 11.0839, 76.9978
Nearby Shelters
1
HAS
Shelter
3.0 km
XYZ, XYZ
24/7
Helps: All Animals
Call Now — 67567968788
Search Again